

Problem

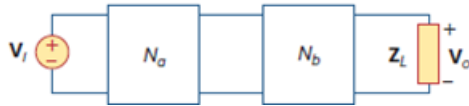
For the individual two-ports shown in Fig. 19.121 where,

$$[z_a] = \begin{bmatrix} 8 & 6 \\ 4 & 5 \end{bmatrix} \Omega \quad [y_b] = \begin{bmatrix} 8 & -4 \\ 2 & 10 \end{bmatrix} \text{ S}$$

(a) Determine the y parameters of the overall two-port.

(b) Find the voltage ratio V_o/V_i when $Z_L = 2 \Omega$.

Figure 19.121



Step-by-step solution

Step 1 of 12

Draw the circuit diagram as shown below.

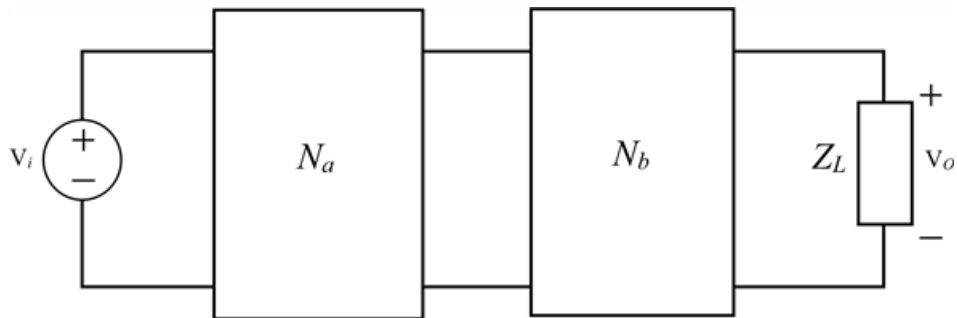


Figure 1

Step 2 of 12

Consider the following data:

$$[z_a] = \begin{bmatrix} 8 & 6 \\ 4 & 5 \end{bmatrix} \Omega$$

$$[y_b] = \begin{bmatrix} 8 & -4 \\ 2 & 10 \end{bmatrix} \text{ S}$$

Step 3 of 12

(a)

Convert z -parameters into T -parameters or **ABCD** parameters.

$$\begin{aligned} [T_a] &= \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \\ &= \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix} \end{aligned}$$

Calculate ΔZ .

$$\begin{aligned} \Delta Z &= (8)(5) - (4)(6) \\ &= 40 - 24 \\ &= 16 \end{aligned}$$

Calculate transmission matrix $[T_a]$.

$$\begin{aligned} [T_a] &= \begin{bmatrix} \frac{8}{4} & \frac{16}{4} \\ \frac{1}{4} & \frac{5}{4} \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 0.25 & 1.25 \end{bmatrix} \end{aligned}$$

Step 4 of 12

Convert y -parameters into T -parameters.

$$[T_b] = \begin{bmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$$

Calculate ΔY .

$$\begin{aligned} \Delta Y &= (8)(10) - (-4)(2) \\ &= 80 + 8 \\ &= 88 \end{aligned}$$

Calculate transmission matrix $[T_b]$.

$$\begin{aligned} [T_b] &= \begin{bmatrix} -\frac{10}{2} & -\frac{1}{2} \\ -\frac{88}{2} & -\frac{8}{2} \end{bmatrix} \\ &= \begin{bmatrix} -5 & -0.5 \\ -44 & -4 \end{bmatrix} \end{aligned}$$

Step 5 of 12

Calculate the overall transmission matrix $[T]$.

$$[T] = [T_a][T_b]$$

Substitute the values of matrices for $[T_a]$ and $[T_b]$ in $[T] = [T_a][T_b]$.

$$\begin{aligned} [T] &= \begin{bmatrix} 2 & 4 \\ 0.25 & 1.25 \end{bmatrix} \begin{bmatrix} -5 & -0.5 \\ -44 & -4 \end{bmatrix} \\ &= \begin{bmatrix} -10 & -176 \\ -1.25 & -55 \end{bmatrix} \\ &= \begin{bmatrix} -186 & -17 \\ -56.25 & -5.125 \end{bmatrix} \end{aligned}$$

Convert the T -parameters to y -parameters.

$$[y] = \begin{bmatrix} \frac{D}{B} & -\frac{\Delta_T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$$

Calculate Δ_T .

$$\begin{aligned} \Delta_T &= (-186)(-5.125) - (-56.25)(-17) \\ &= 953.25 - 956.25 \\ &= -3 \end{aligned}$$

Step 6 of 12

Calculate y -parameters $[y]$.

$$\begin{aligned} [y] &= \begin{bmatrix} \frac{-5.125}{-17} & -\frac{(-3)}{(-17)} \\ -\frac{1}{(-17)} & \frac{-186}{-17} \end{bmatrix} \\ &= \begin{bmatrix} 0.3015 & -0.1765 \\ -0.0588 & 10.94 \end{bmatrix} \end{aligned}$$

Therefore, the overall y -parameters is $\begin{bmatrix} 0.3015 & -0.1765 \\ -0.0588 & 10.94 \end{bmatrix}$.

Step 7 of 12

(b)

Convert the T -parameters to z -parameters.

$$\begin{aligned} [z] &= \begin{bmatrix} \frac{A}{C} & \frac{\Delta_T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix} \\ &= \begin{bmatrix} \frac{-186}{-56.25} & \frac{-3}{-56.25} \\ \frac{1}{-56.25} & \frac{-5.125}{-56.25} \end{bmatrix} \\ &= \begin{bmatrix} 3.3067 & 0.0533 \\ -0.0178 & 0.0911 \end{bmatrix} \end{aligned}$$

Step 8 of 12

Draw the equivalent circuit diagram as shown below.

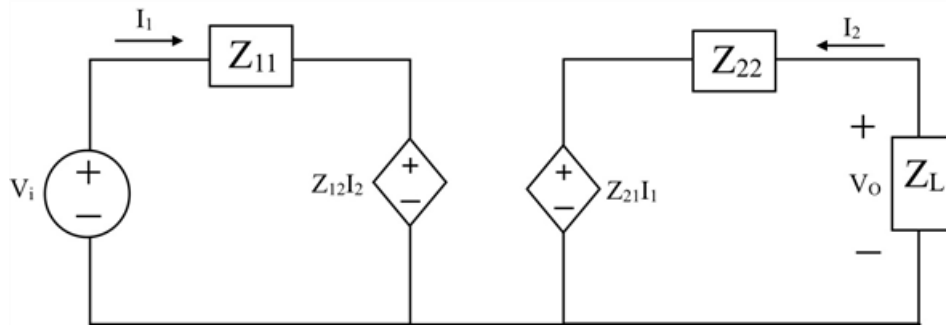


Figure 2

Step 9 of 12

From Figure 2, apply Kirchhoff's voltage law to left loop.

$$-V_i + Z_{11}I_1 + Z_{12}I_2 = 0$$

$$V_i = Z_{11}I_1 + Z_{12}I_2 \quad \dots\dots (1)$$

From Figure 2, apply Kirchhoff's voltage law to right loop.

$$-V_o + Z_{22}I_2 + Z_{21}I_1 = 0$$

$$V_o = Z_{22}I_2 + Z_{21}I_1 \quad \dots\dots (2)$$

Step 10 of 12

From Figure 2, write the voltage across load.

$$V_o = -I_2 Z_L$$

$$I_2 = -\frac{V_o}{Z_L} \quad \dots\dots (3)$$

Substitute $-\frac{V_o}{Z_L}$ for I_2 in $V_o = Z_{22}I_2 + Z_{21}I_1$.

$$V_o = (Z_{22})\left(-\frac{V_o}{Z_L}\right) + Z_{21}I_1$$

$$Z_{21}I_1 = V_o + (Z_{22})\left(\frac{V_o}{Z_L}\right)$$

$$I_1 = V_o \left(\frac{1}{Z_{21}} + \frac{Z_{22}}{Z_{21}Z_L} \right) \quad \dots\dots (4)$$

Substitute $-\frac{V_o}{Z_L}$ for I_2 and $V_o \left(\frac{1}{Z_{21}} + \frac{Z_{22}}{Z_{21}Z_L} \right)$ for I_1 in $V_i = Z_{11}I_1 + Z_{12}I_2$.

$$V_i = Z_{11} \left(V_o \left(\frac{1}{Z_{21}} + \frac{Z_{22}}{Z_{21}Z_L} \right) \right) + Z_{12} \left(-\frac{V_o}{Z_L} \right)$$

$$= V_o \left(\frac{Z_{11}}{Z_{21}} + \frac{Z_{11}Z_{22}}{Z_{21}Z_L} - \frac{Z_{12}}{Z_L} \right)$$

$$\frac{V_i}{V_o} = \frac{Z_{11}}{Z_{21}} + \frac{Z_{11}Z_{22}}{Z_{21}Z_L} - \frac{Z_{12}}{Z_L}$$

Step 11 of 12

Substitute 3.3067 for Z_{11} , 0.0533 for Z_{12} , -0.0178 for Z_{21} , 0.0911 for Z_{22} and 2 for Z_L in

$$\frac{V_i}{V_o} = \frac{Z_{11}}{Z_{21}} + \frac{Z_{11}Z_{22}}{Z_{21}Z_L} - \frac{Z_{12}}{Z_L}.$$

$$\begin{aligned}\frac{V_i}{V_o} &= \frac{3.3067}{-0.0178} + \frac{(3.3067)(0.0911)}{(-0.0178)(2)} - \frac{0.0533}{2} \\ &= -185.77 - 8.46 - 0.0267 \\ &= -194.3\end{aligned}$$

Step 12 of 12

Calculate $\frac{V_o}{V_i}$.

$$\begin{aligned}\frac{V_o}{V_i} &= -\frac{1}{194.3} \\ &= -0.0051\end{aligned}$$

Therefore, the voltage ratio $\frac{V_o}{V_i}$ is -0.0051.